

KCT ENGINEERING COLLEGE, KALABURAGI

SUBJECT: Introduction to Electrical Engineering SUB CODE: BESCK104B/204B

QUESTION BANK

MODULE-1: Introduction, Power Generation, DC Circuits

1. Differentiate between conventional & non-conventional energy sources.
2. With a neat single line diagram explain the various steps of electrical power transmission and distribution.
3. With block diagram explain nuclear power generation.
4. With block diagram explain Hydel power generation.
5. With block diagram explain solar power generation.
6. With block diagram explain wind power generation.
7. State & explain Ohm's law with its limitations.
8. State & explain Kirchhoff's laws.

MODULE-2: AC Fundamentals, Three Phase Circuits

1. Define the following terms (a) Power, (b) Energy, (c) Frequency (d) Time period (e) average value (f) rms value (g) form factor (h) peak factor (i) power factor
2. Explain the phasor relationship between voltage & current when an AC is passed through a pure Resistor.
3. Explain the phasor relationship between voltage & current when an AC is passed through a pure Inductor. Show that average power consumed is zero.
4. Explain the phasor relationship between voltage & current when an AC is passed through a pure capacitor. Show that average power consumed is zero.
5. Explain the behavior of the series RL circuit. Draw its phasor diagram.
6. Explain the behavior of the series RC circuit. Draw its phasor diagram.
7. Define power factor. Explain its significance in ac circuit.
8. Derive an expression for impedance, phase angle & power for series RLC circuit energized by sinusoidal voltage.
9. Explain the generation of three-phase voltages.
10. List the advantages of a three-phase system over a single-phase system.
11. Obtain the relationship between line & phase values in a three-phase balanced star connected system.
12. Obtain the relationship between line & phase values in a three-phase balanced delta connected system.

MODULE-3: DC Machines, DC Generator, DC Motor

1. Explain the working principle of the DC generator.
2. With a neat sketch explain the construction of the DC machine.
3. Derive the emf equation of the DC generator.
4. Explain the various types of DC generators.
5. State the applications of various types of DC generators.
6. Explain the working principle of the DC motor.
7. What is back emf? State its significance.

8. Derive the expression for torque developed in a DC motor.
9. Explain the various characteristics of DC motor.
10. Write the applications of DC motors.

MODULE-4: Transformers, Three-Phase Induction Motors

1. Explain the principle of operation of single-phase transformers
2. With a neat sketch explain the constructional details of core & shell-type transformers.
3. Derive the emf equation of a transformer
4. Explain the various losses in a transformer
5. Define the efficiency of a transformer? Derive the condition for the maximum efficiency for a transformer.
6. What is a rotating field? What is the speed of a rotating magnetic field?
7. Explain the principle of operation of induction motor.
8. Discuss the important features of squirrel cage & phase wound rotor type induction motor.
9. Derive the expression for slip & frequency of rotor currents.
10. Explain slip and its significance.

MODULE-5: Domestic Wiring, Electricity Bill, Equipment Safety Measures Personal Safety Measures

1. Explain the requirements of domestic wiring.
2. With neat diagram explain casing and capping type of wiring.
3. With neat diagram and switching table explain two way control of lamps.
4. With neat diagram and switching table explain three way control of lamps.
5. Discuss how electricity bill is calculated based on "unit" which is consumption of electrical energy for domestic consumers.
6. Explain two part tariff with its advantages & disadvantages.
7. List out the power rating of household appliances including air conditioners, PCs, laptops, printers, etc. Find the total power consumed.
8. With a neat circuit diagram, explain the operation of Fuse, MCB and RCCB.
9. What is earthing? Why earthing is required?
10. With the help of neat sketch, explain plate earthing and pipe earthing.
11. Write a short note on precautions against an electric shock.